

PAX-TEC Interface Document; March 17, 2004

Power supplies

SETI PROVIDED

+15 Volt TEC -0/+10%: 8 A max.
-15 Volt +/-5%: 50 mA max

All the supplies will be linear type regulators.

PSI PROVIDED

Power supply pigtails: # 22 stranded wire; 15" long; color code:

Red +15 Volt TEC 4ea
Black ground (+15 Volt TEC return) 4ea
Blue -15 Volt 1ea
Brown ground (-15 Volt return)

Connections between TEC Driver Amps and TECs

Red +15 Volt TEC 4ea, #22 stranded wire, 15 ft max
Black ground (+15 Volt TEC return) 4ea, #22 stranded wire, 15 ft max

Mechanical

Dimensions:

PAX: 11.5" × 13.0" × 2.25" box with 1/4" and 3/8" aluminum walls

PAX-TEC heatsink dimensions (Mounted on back of PAX): 11.5" × 13.0" × 1.8" max
Mounting holes for PAX-TEC heatsink TBD

TEC Driver Amp heatsink: 800 sq. inches surface area max

TEC Driver Amp volume: 40 cu. inches

Thermal grease to be used between all thermally conducting paths



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Thermal

Maintain PAX base-plate temperature at 20 ± 3 C

PAX internal heat load: 10 W (8W present estimate)

Cooling air for PAX-TEC heatsink, TEC Driver Amps and TEC Driver Amp heatsinks: 15 CFM
min @ 30 Deg. C max

Ambient: PAX Box: -29 to 43.5 C